

Sta313 lec0201

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Assignment 1

Due Date: February 6, 2026

The Level of Happiness in Winter: Tracking a University of Toronto Student's Survival Across Five Workdays—Number of Layers, Social Context, and Energy Drain

Dataset:

	Date	Snapshot time (Morning=11:59; Afternoon=17:59; Evening=23:59)	Activity (Sleeping/Studying/Transit/Leisure)	Energy Drain
	Monday	Morning	Studying	Medium
	Monday	Afternoon	Leisure	Low
	Monday	Evening	Studying	Medium
	Tuesday	Morning	Sleeping	Low
	Tuesday	Afternoon	Studying	Medium
	Tuesday	Evening	Sleeping	Low
	Wednesday	Morning	Transit	High
	Wednesday	Afternoon	Studying	Medium
	Wednesday	Evening	Sleeping	Low
	Thursday	Morning	Studying	Medium
	Thursday	Afternoon	Transit	High
	Thursday	Evening	Studying	Medium
	Friday	Morning	Studying	Medium
	Friday	Afternoon	Leisure	Low
	Friday	Evening	Studying	Medium
Data Type:	Cyclic Ordinal	Cyclic Ordinal	Nominal Categorical	Sequential Ordinal

Location(Home/Classroom/Outdoor)	Body Temperature (Cold / Comfortable / Hot)	Number of Layers (Top layers + Bottom layers)	Social Context (With Friends/Alone)	Happiness Levels (1-5)
Classroom	Hot	4	With Friends	3
Home	Comfortable	2	Alone	4
Home	Comfortable	1	Alone	2
Home	Comfortable	1	Alone	5
Home	Comfortable	2	Alone	2
Home	Comfortable	1	Alone	5
Outdoor	Cold	5	Alone	1
Classroom	Hot	4	With Friends	3
Home	Comfortable	1	Alone	5
Home	Comfortable	2	With Friends	3
Outdoor	Cold	5	Alone	1
Home	Comfortable	2	Alone	2
Classroom	Hot	4	With Friends	3
Home	Comfortable	2	With Friends	4
Home	Comfortable	1	Alone	2
Nominal Categorical	Diverging Ordinal	Sequential Quantitative	Nominal Categorical	Diverging Ordinal

Explanation:

Date (Cyclic Ordinal): I decided to gather data during the third week of the winter semester, which is January 19–23. I selected this period because it signifies the stabilization of my academic schedule after the deadline for course enrollment. This guarantees that the data represents a typical pattern rather than the anomalies of the beginning of the semester.

Snapshot time (Cyclic Ordinal): I selected three fixed timestamps (11:59, 17:59, and 23:59) as discrete sampling points that capture the dominant states of the morning, afternoon, and evening. Recording at consistent, end-of-period moments helps reduce short-lived fluctuations that might occur mid-activity and provides a more stable snapshot of each time block. This makes it easier to compare typical daily patterns across time periods.

Activity (Nominal Categorical): The specific activity I was engaged in at the snapshot moment. I set recurring digital alarms at each designated time; when an alarm went off, I recorded my current activity note on my phone.

Energy Drain (Sequential Ordinal): My perceived level of physical and mental depletion at the snapshot moment. Self-rated at each alarm using three levels (Low/Medium/High).

Location (Nominal Categorical): Where I was at the snapshot moment (e.g., home, classroom, outside). Recorded at each alarm based on where I was physically located.

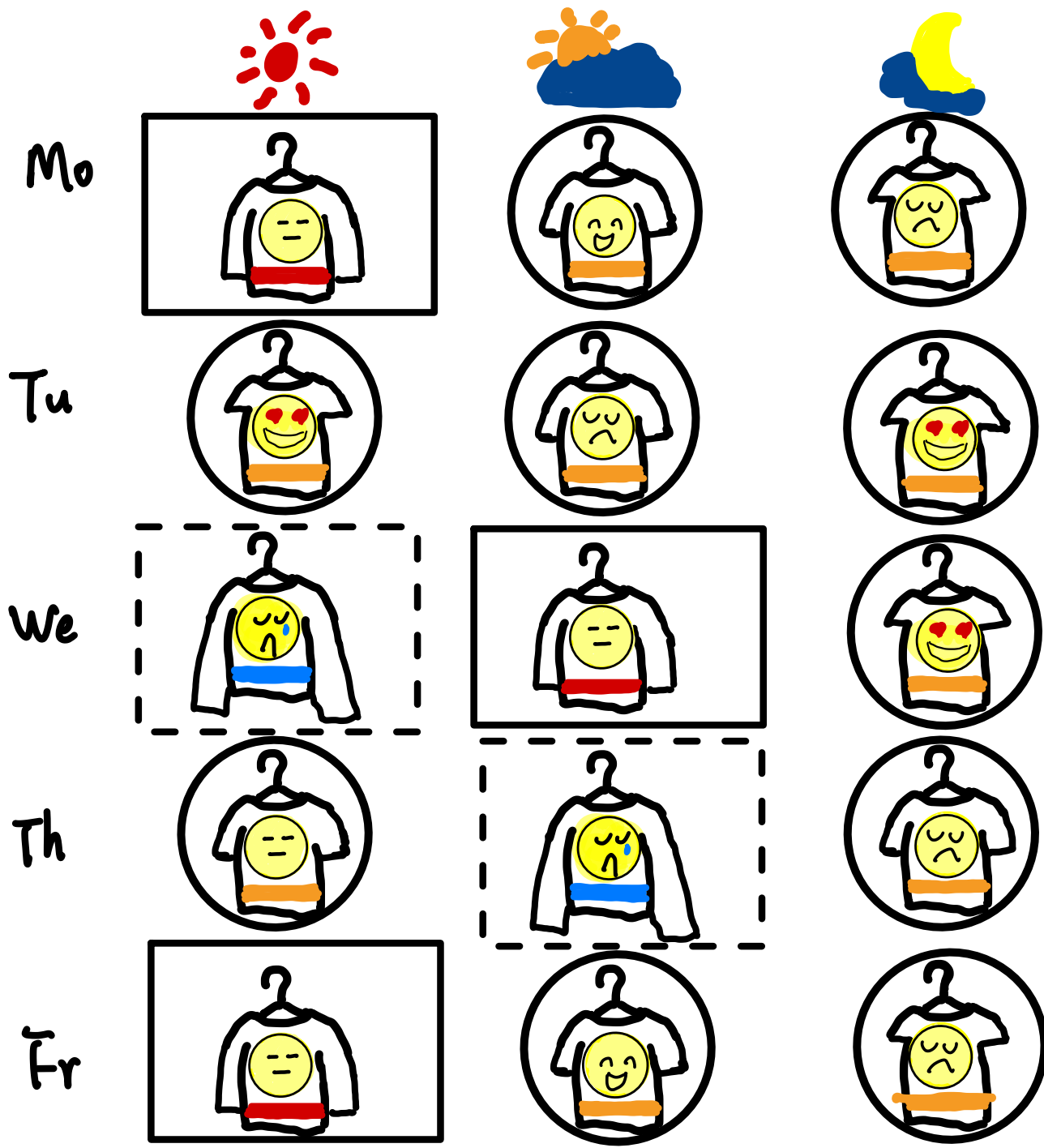
Body Temperature (Diverging Ordinal): My subjective thermal sensation relative to the immediate environment. Self-reported at each alarm using three levels (Cold/Comfortable = Midpoint/Hot).

Number of Layers (Sequential Quantitative): The amount of clothing worn at the snapshot moment. Manually counted the total number of clothing items worn (upper + lower body) at each alarm, with 0 indicating no additional layers.

Social Context (Nominal Categorical): My immediate social setting at the snapshot moment (e.g., Alone vs. With friends). Recorded at each alarm based on direct observation of who I was with.

Happiness Levels (Diverging Ordinal): My self-assessed emotional well-being at the snapshot moment. Self-rated at each alarm on a 1–5 scale (1 = lowest, 3 = neutral, 5 = highest).

The cost of warmth: How layering up weighed down my happiness?



Legend

Number of layers

1
2
3
4
5

Happiness Levels

1
2
3
4
5

Location

Home
Classroom
Outside

Body Temperature

Hot
comfortable
Cold

Snapshot time

Morning (11:59)
Afternoon (17:59)
Evening (23:59)

Date

Monday Mo
Tuesday Tu
Wednesday We
Thursday Th
Friday Fr

Description of Sketch 1 (398 words):

In the first sketch, I aim to investigate the relationship between emotional well-being and physical thermal comfort across three environments in my study life. The sketch highlights a daily trade-off: staying warm comes at the cost of comfort and mobility.

Firstly, I organized the canvas as a temporal grid using spatial position as the primary visual channel. Columns encode “snapshot time” as cyclic ordinal (morning/afternoon/evening), and rows encode “date” as cyclic ordinal (Mon–Fri). This layout enables quick comparison through position on a common scale (magnitude channel) and is supported by the **principle of proximity**. Header icons (sun/cloud/moon) act as **pre-attentive cues** to distinguish time of day.

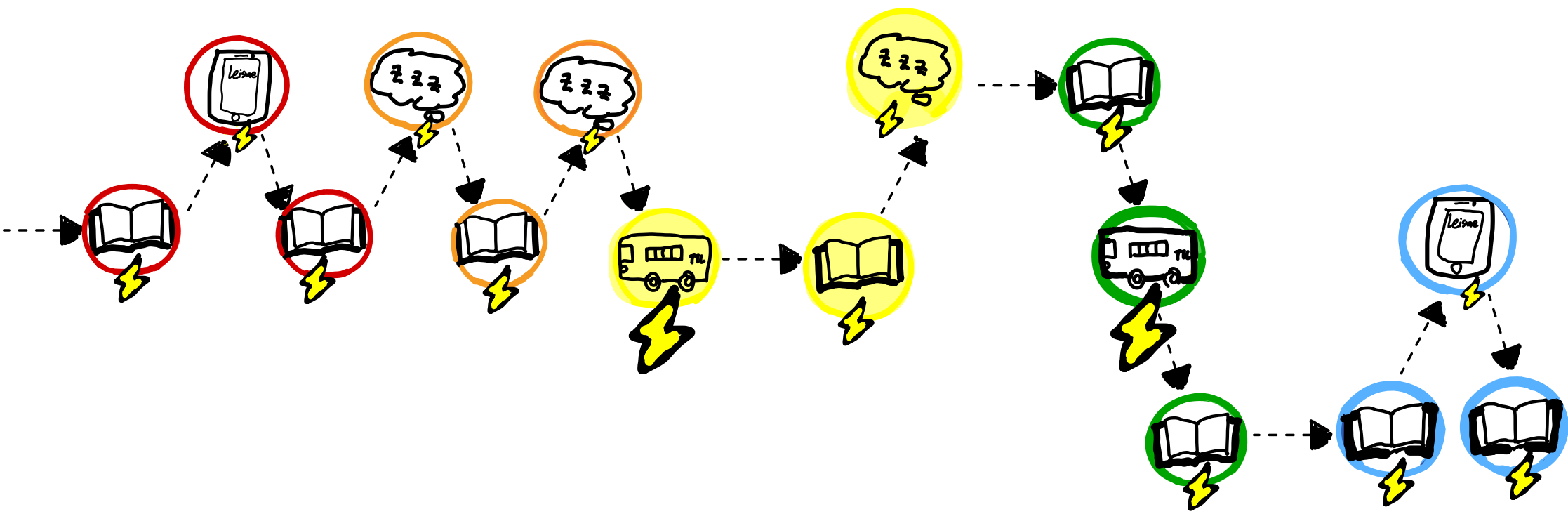
Within each grid cell, I encoded the categorical attribute “location” using three distinct container shapes (identity channel). I repeat the same container shape for each location, using **similarity** to make location-based groups easy to spot. For “outside,” I used a dashed boundary to suggest openness while still defining a clear region; viewers perceptually complete the broken contour through **closure**, reading it as one boundary rather than disconnected segments. **Enclosure** bundles the temperature bar, clothing icon, and facial expression into one observation, so each cell is readable on first viewing.

Alongside location, I represented “body temperature” by drawing a colored bar directly beneath the clothing icon within each cell. Further, body temperature is a three-level diverging ordinal variable (cold/comfortable/hot), centered on a neutral midpoint. I mapped it to a diverging hue scheme (blue–yellow–red): the opponent hues (blue vs. red) emphasize the direction of deviation from comfort, while yellow marks the baseline state. These common temperature conventions enable viewers to track shifts in thermal comfort across locations instantly.

Regarding the inner icons, I mapped the “number of layers” as a sequential quantitative attribute using sleeve length (magnitude channel): longer sleeves indicate more layers. This reduces cognitive load by making layering easy to read without repeatedly checking the legend. For “happiness levels”, I encoded a 5-point diverging ordinal scale using facial expressions, varying the mouth curvature (magnitude channel) from a frown to a smile to support immediate ordinal judgment of emotion.

I prioritized intuitive readability over precision. For layering, I didn’t label exact layer counts and instead used sleeve length to keep the icons clean. For body temperature, I relied on familiar hues convention for quick recognition, even though luminance or saturation would align more strictly with magnitude channels for ordered values.

Mapping the Impact of Energy Drain on Emotions During Workdays



Legend	Activity	Date	Wednesday	Happiness Levels
Energy Drain				
Low ⚡	Study 📖	Monday 🔴		1-2 sad ↘
Medium ⚡	Leisure 📱	Tuesday 🟠		3 so-so →
High ⚡	Transit 🚌	Mon → Fri		4-5 happy ↗
	Sleeping ☁️		Thursday 🟢	
			Friday 🔵	

Description of Sketch 2 (400 words):

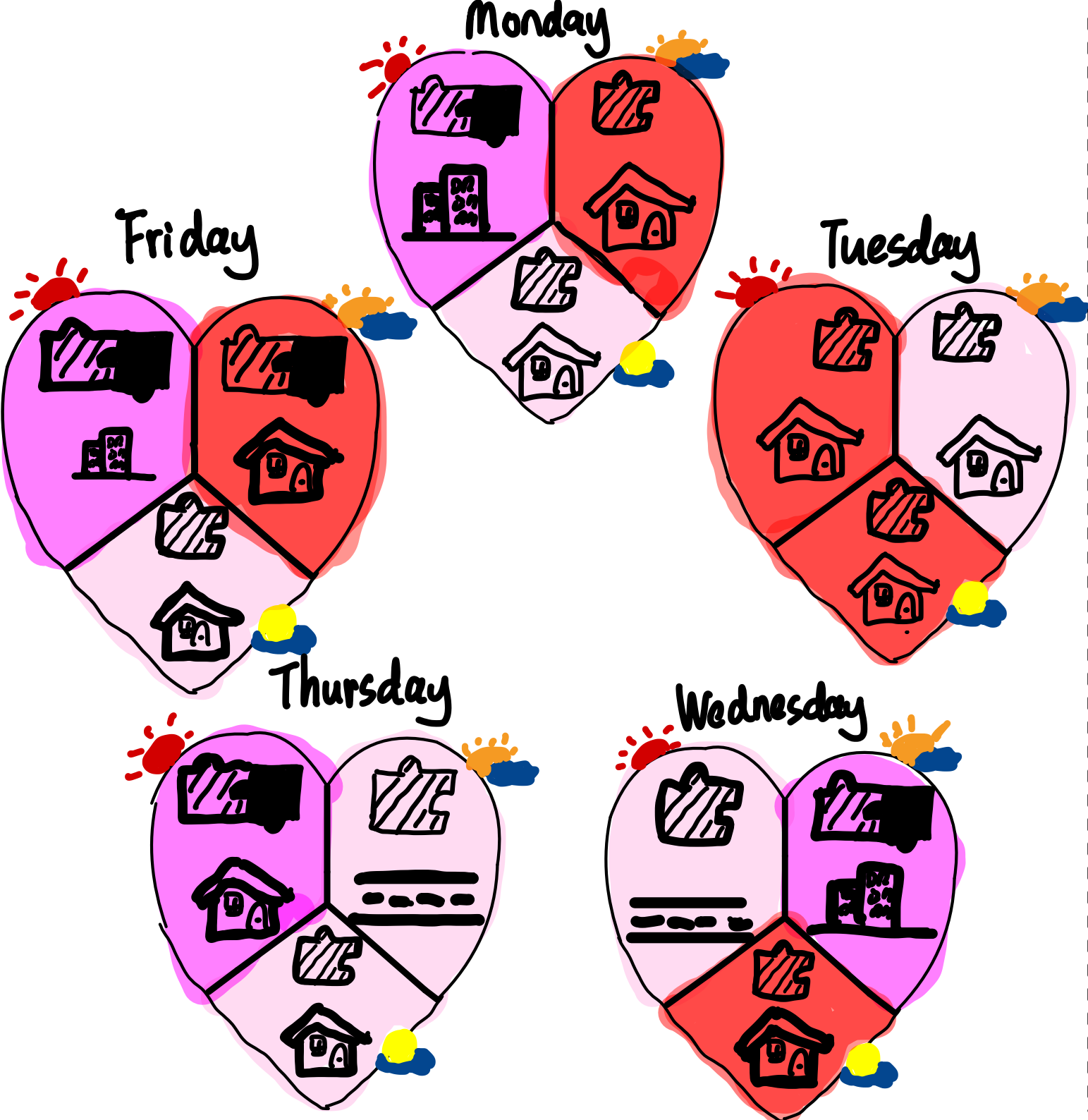
In the second sketch, I aim to visualize how energy drain from my daily routine at the University of Toronto directly impacts my emotional state. I treated my energy levels as a “mental journey” to capture how fatigue accumulates over time. This approach helps show the real-time ebb and flow of my emotions and highlights how exhaustion from one day often carries over into the next.

I used **connectedness** to join observations into a continuous path with right-pointing arrows, which sets the reading order from left to right and makes the timeline progression explicit. The arrows also encode the diverging ordinal variable “happiness level”: I grouped the 5-point scale into three bands (1–2 sad, 3 so-so, 4–5 happy). Visually, I used only the tilt/angle of the arrows (magnitude channel) to communicate emotional direction: sad arrows tilt down-right, so-so arrows point straight right, and happy arrows tilt up-right. To avoid mixing intensity with other visual cues, I kept the line and arrow lengths constant, so viewers focus on changes in direction and can easily follow how the mood shifts over time. In particular, the sequence shows a clear drop on Thursday, and that fatigue carries into Friday.

To encode the cyclic ordinal variable “date”, I drew the same-size circular outlines and mapped each day to a distinct color hue (identity channel). Even though “date” is a cyclic ordinal, I used hue for day identity rather than ordering, since the connected path already encodes the reading order. I assigned each day a consistent hue, applying **similarity** to group its observations. I then use **proximity** by clustering each day’s points, which visually segments the path into day-level blocks for quick comparison across days.

“Activity” is a nominal categorical attribute, so I encode it with intuitive activity icons (identity channel) to support quick recognition. “Energy drain” is a three-level sequential ordinal variable (low/medium/high), which I map to the size of a lightning symbol (magnitude channel): a larger lightning icon indicates higher depletion, making differences in intensity easy to compare.

In general, I chose clarity and storytelling over fine-grained precision in happiness. Instead of five subtle angles, I grouped the scale into three bands, which makes the sketch easier to read and highlights the carryover fatigue from Thursday into Friday. Moreover, “date” is distinguished by color hue, which is less accessible for color-vision deficiencies; a small day label would be a clearer alternative.



My Heart Containers

During workdays: Emotions Across Social contexts

Legend

Snapshot time

Morning (11:59)

Afternoon (17:59)

Evening (23:59)



Social context

Alone

With Friends



Happiness Levels

1-2 Sad

3 So-so

4-5 Happy

Location

Home

classroom

Outside



Description of Sketch 3 (393 words):

In the third sketch, I track my workday moods at the University of Toronto alongside social context and location to see which environments amplify or buffer the emotional impact of social interaction. I want to determine my optimal "social rhythm" and understand what kind of social contexts and places contribute most effectively to my happiness levels.

In the layout, I designed each workday as its own heart-shaped container, because a heart has an intuitive association with emotion and signals the theme without extra explanation. The heart boundary applies **enclosure** that elements inside the same heart are naturally interpreted as data from the same day. Within each heart, I further divide the space into three observation points to encode "snapshot time" (morning/afternoon/evening), arranged clockwise to create a cyclical reading path within the day. Moreover, I also include three clear time-of-day symbols as **pre-attentive cues** for easy recognition. Meanwhile, I designed the five hearts in a circular ring from Monday to Friday, presenting the workweek as a repeating sequence and spatially echoing the cyclic ordinal nature of both "date" and "snapshot time".

"Happiness level" is a diverging ordinal variable. To make each workday easier to compare, I grouped the original 5-point ratings into three bands (1–2 sad, 3 so-so, 4–5 happy). This reduces visual clutter since various colors would be harder to distinguish at a small scale. Then, I mapped the bands to color saturation (magnitude channel), using a progressively more saturated red to indicate a more positive mood; in other words, the redder the mark, the better I felt that day.

"Location" is a nominal categorical variable, so I encoded it with intuitive place icons (identity channel), since icons stay readable at small scales. "Social context" is also nominal categorical, so I used a puzzle-piece metaphor where a single piece, while with friends is two interlocking pieces. This design makes social connections immediately visible, and repeating the same puzzle forms applies **similarity** to group observations with the same social context.

I encoded happiness with color saturation, but color is a lower-ranked channel for ordered values and can be hard to read for color-vision deficiencies. To improve discriminability at a small scale, I reduced cardinality by collapsing the 5-point ratings into three bands (sad/so-so/happy). This sacrifices nuance, but makes within-day snapshot comparisons clearer. A possible alternative is adding a texture or label cue.

Reflection (300 words):

These sketches demonstrate how the number of layers, energy drain of various activities, and social context influence my winter-term workday mood at the University of Toronto. Sketch 1 prioritizes structured comparison, Sketch 2 prioritizes narrative flow, and Sketch 3 prioritizes encoding interpretation.

The strength of Sketch 1 is that the day and snapshot grid make the structure unambiguous. Moreover, its five-level happiness scale also preserves fine emotional nuance. Compared with Sketches 2 and 3, each cell of Sketch 1 contains more marks, which reduces separability and sends viewers back to the legend. To improve discriminability, the next step is to simplify the clothing icon and move body temperature to a separate small thermometer/bar.

Sketch 2 is more narrative than Sketch 1 because connectedness turns the week into a continuous path, making mood transitions and fatigue carryover feel like a time-progressing process. Compared with Sketch 1's five-level scale, it compresses happiness from five to three bands since subtle angles are hard to read at a small scale and relies on color, which is less accessible for color-vision deficiencies. As a follow-up, I can differentiate days with outline styles (solid/dashed/dotted) and/or insert small gaps between day segments.

Sketch 3 is the most engaging and memorable. Compared with Sketch 2, the puzzle-piece metaphor makes social context faster to scan and more distinctive. It also collapses happiness into three bands to improve readability at small sizes. Nevertheless, Sketch 3 shares Sketch 2's weakness, which is that ordered happiness is still mainly conveyed by color, a weaker channel for ranking, and not colorblind-friendly. A consistent upgrade for both Sketch 2 and Sketch 3 is redundant non-color encoding (simple line patterns inside hearts) to keep the mood readable without clutter.

Overall, Sketch 1 best supports exact comparisons, while Sketches 2 and 3 better convey lived progression.

AI Help:

I used ChatGPT to suggest alternative expressions for “at a glance” (e.g., “immediately,” “on first viewing,” “instantly”) so the description reads more smoothly and avoids repetitive wording.